

Robotics 2

Hybrid Force/Motion Control

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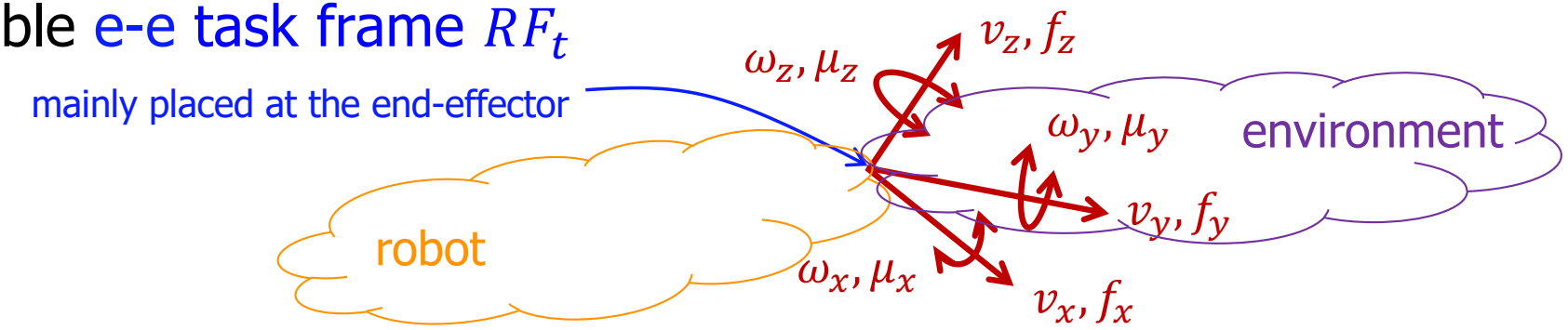
Hybrid force/motion control



- we consider contacts/interactions between the robot and a stiff environment that naturally constrains the end-effector (e-e) motion
- compared to an approach using the constrained/reduced robot dynamics with (bilateral) geometric constraints, the differences are
  - the hybrid control law is designed in ideal conditions, but now unconstrained directions of motion and constrained force directions are defined in a more direct way using a task frame formalism
  - all non-ideal conditions (compliant surfaces, friction at the contact, errors in contact surface orientation) are handled explicitly in the control scheme by a geometric filtering of the measured quantities
    - only signal components that should appear in certain directions based on the nominal task model are considered
    - those that should not be there are treated as disturbances to be rejected
- in this way, the hybrid control law avoids to introduce conflicting behaviors (force vs. motion control) in each task space direction!

Natural constraints

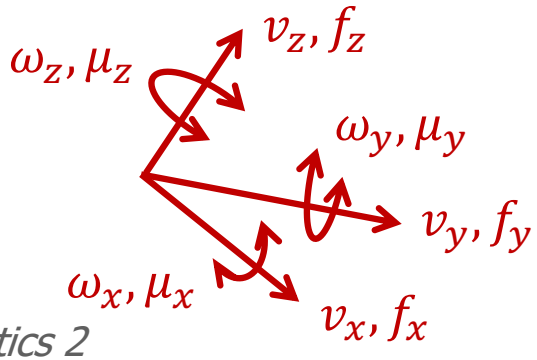
- in ideal conditions (robot and environment are perfectly rigid, contact is frictionless), two sets of generalized directions can be defined in a 6-dimensional end-effector (e-e) task space
  - end-effector velocities (v/ω) are prohibited along/around k directions (since the environment reacts there with forces/moments)
  - reaction forces/moments (f/μ) are absent along/around $6-k$ directions (where environment does not prevent end-effector motion)
- these constraints are usually called natural constraints on motion and force associated to the geometry of robot-environment contact
- the two sets of directions are characterized through the axes of a suitable e-e task frame RF_t



Artificial constraints



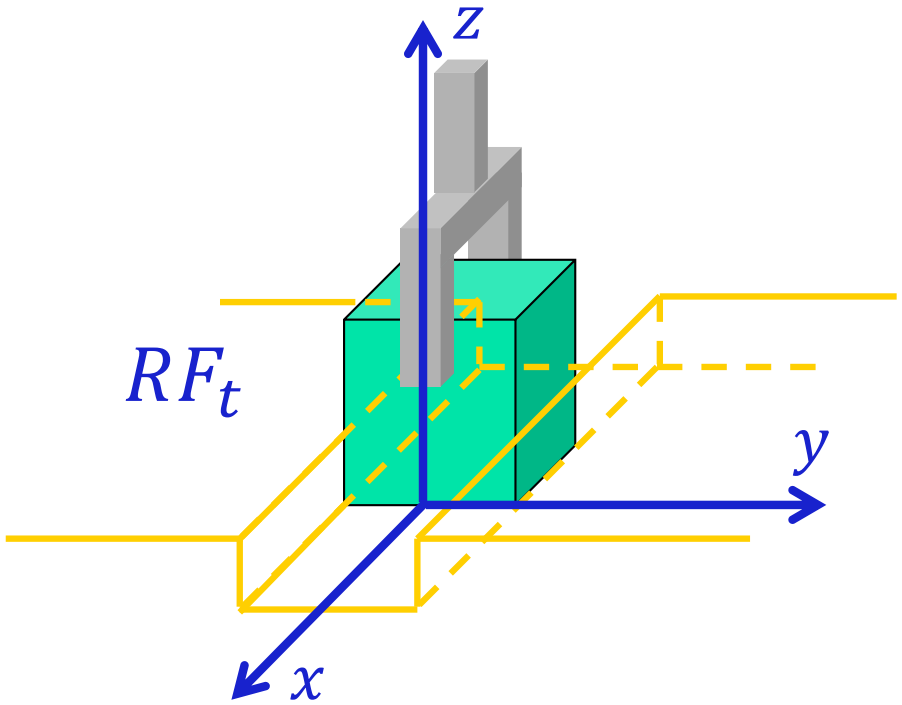
- the way task execution should be performed can be expressed in terms of so-called artificial constraints that specify the desired values (to be imposed by the control law)
  - for the end-effector velocities (v/ω) along/around $6-k$ directions where free motion can occur
  - for the contact forces/moments (f/μ) along/around k directions where an admissible reaction of the environment can occur
- the two sets of directions are complementary (they cover the 6D generalized task space) and mutually orthogonal, while the task frame can be time-varying ("moves with task progress")
  - "directions" are intended as 6D e-e screws: twists $t_e = (v^T \ \omega^T)^T$ and wrenches $w_e = (f^T \ \mu^T)^T$



$$w_e{}^T t_e = 0 \ \Leftrightarrow \ \text{orthogonality}$$

but ill-defined (don't use it!) for $t_{e1}^T t_{e2}$ or $w_{e1}^T w_{e2}$

Task frame and constraints - example 1



task: slide the cube along a guide

natural (geometric) constraints

$v_y = v_z = 0$
$\omega_x = \omega_z = 0$ } $k = 4$

$f_x = \mu_y = 0$ } $6 - k = 2$

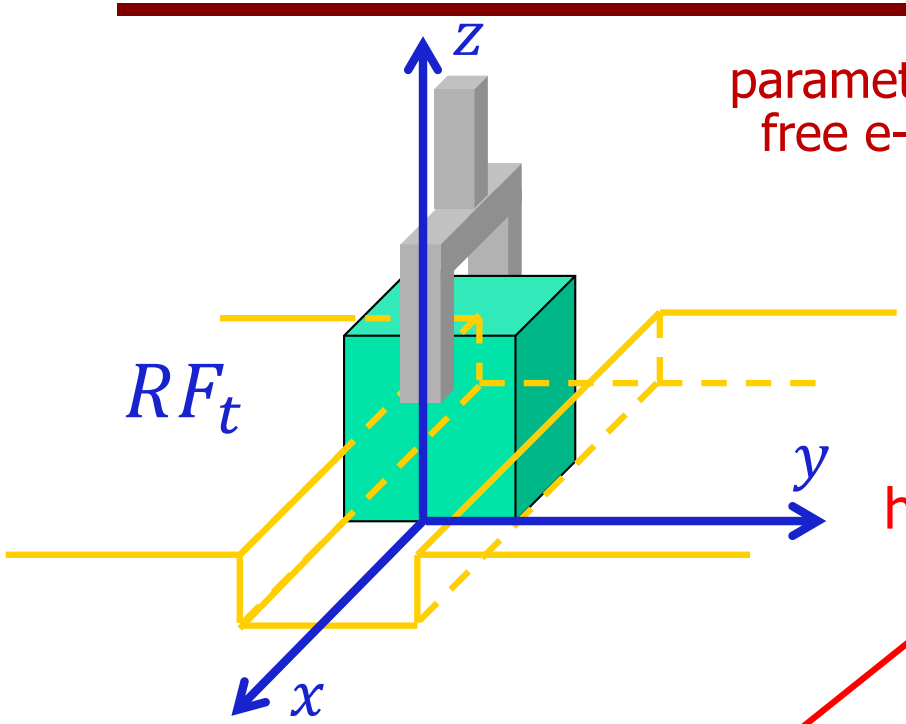
artificial constraints
(to be imposed by the control law)

$k = 4$ {
- $f_y = f_{y,des} \ (= 0)$ (to avoid internal stress)
- $\mu_x = \mu_{x,des} \ (= 0), \mu_z = \mu_{z,des} (= 0)$
- $f_z = f_{z,des}$ (to keep contact)

$6 - k = 2$ {
- $\omega_y = \omega_{y,des} = 0$ (to slide and not to roll !!)
- $v_x = v_{x,des}$

v = velocity
ω = angular velocity
f = force
μ = moment

Selection of directions - example 1



parametrization of free e-e motion

$$t_e = \begin{pmatrix} v \\ \omega \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} v_x \\ \omega_y \end{pmatrix} = S_t \, v$$

selected twist components

here, constant and unitary ("selection" of columns from the 6×6 identity matrix)

parametrization of admissible reaction of the environment

$$w_{\text{env}} = \begin{pmatrix} f \\ \mu \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} f_y \\ f_z \\ \mu_x \\ \mu_z \end{pmatrix} = S_w \, \lambda$$

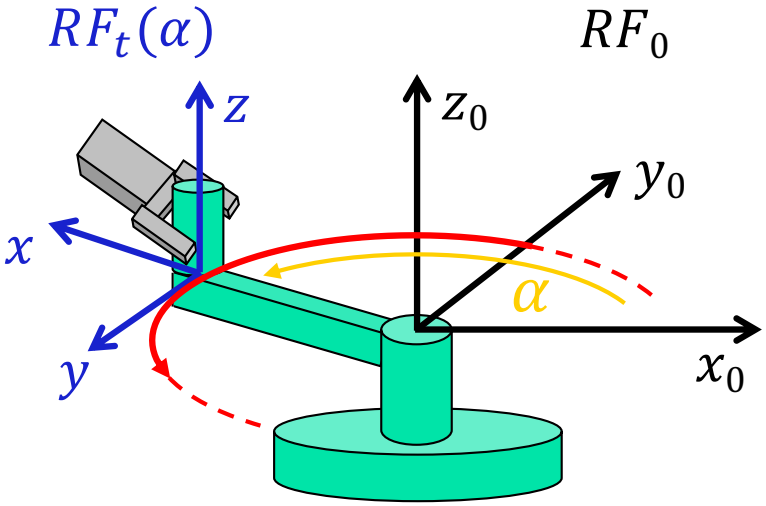
selected wrench components

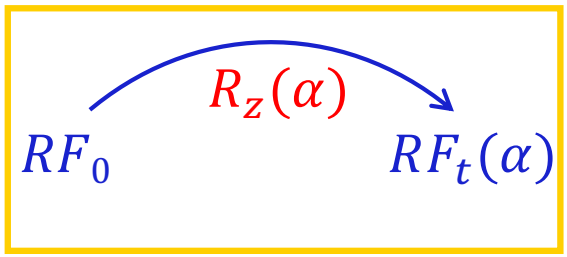
$$S_w^T \, S_t = 0$$

reaction forces/torques do **not** perform work on feasible motions

$$\begin{pmatrix} f^T & \mu^T \end{pmatrix} \begin{pmatrix} v \\ \omega \end{pmatrix} = 0$$

Task frame and constraints - example 2





task: turning a crank (free handle)

natural constraints

$v_x = v_z = 0$

$\omega_x = \omega_y = 0$

$f_y = \mu_z = 0$

artificial constraints

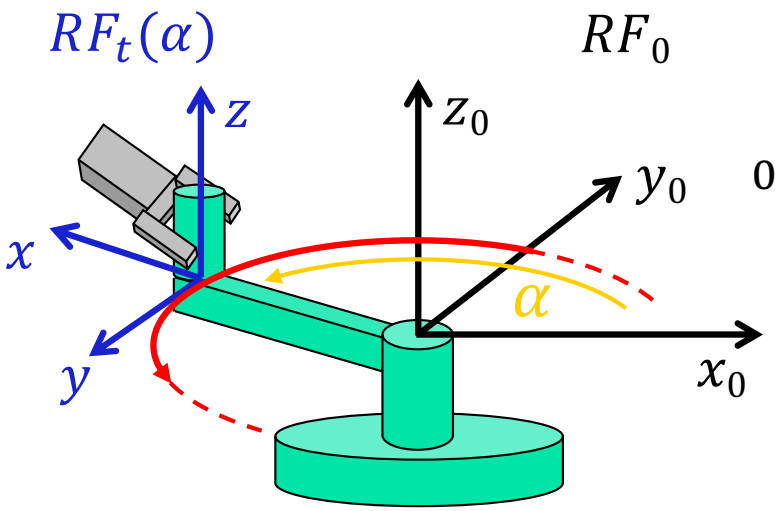
$f_x = f_{x,des}\ (= 0), f_z = f_{z,des}\ (= 0)$

$\mu_x = \mu_{x,des}\ (= 0), \mu_y = \mu_{y,des}\ (= 0)$

$v_y = v_{y,des}$ (the tangent speed of rotation)

$\omega_z = \omega_{z,des}$ ($= 0$ if handle should not spin)

Selection of directions – example 2



parametrization of feasible motions

$$ {}^0t_e = \begin{pmatrix} {}^0v \\ {}^0\omega \end{pmatrix} = \begin{pmatrix} R^T(\alpha) & 0 \\ 0 & R^T(\alpha) \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} v_y \\ \omega_z \end{pmatrix} $$

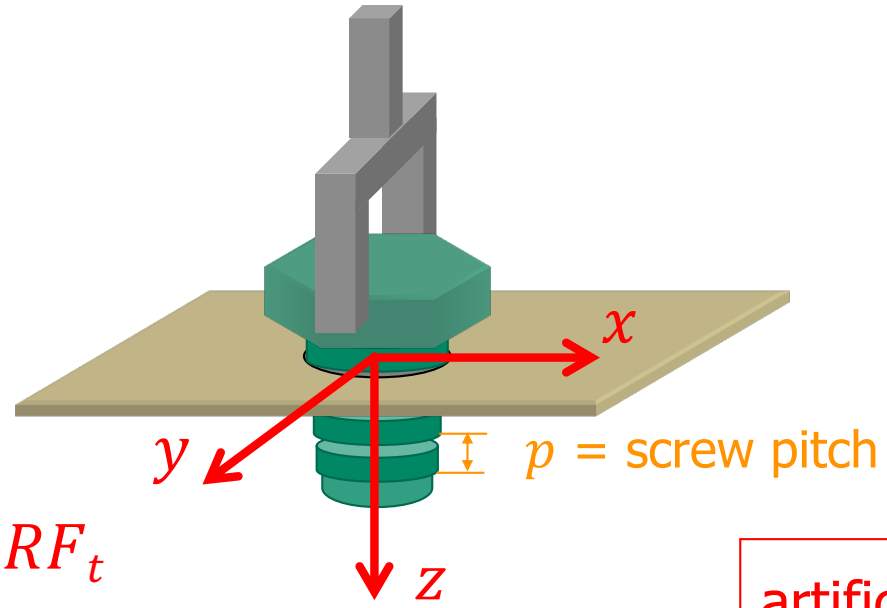
$$ = S_t(\alpha)\, v $$

$$ S_w^T(\alpha)\, S_t(\alpha) = 0 $$

parametrization of feasible reactions

$$ {}^0w_{\text{env}} = \begin{pmatrix} {}^0f \\ {}^0\mu \end{pmatrix} = \begin{pmatrix} R^T(\alpha) & 0 \\ 0 & R^T(\alpha) \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} f_x \\ f_z \\ \mu_x \\ \mu_y \end{pmatrix} = S_w(\alpha)\, \lambda $$

Task frame and constraints - example 3



task: insert a screw into a bolt

natural constraints (partial…)

$v_x = v_y = 0$

$\omega_x = \omega_y = 0$

artificial constraints (abundant…)

$f_x = f_{x,des} = 0, f_y = f_{y,des} = 0$

$\mu_x = \mu_{x,des} = 0, \mu_y = \mu_{y,des} = 0$

$v_z = v_{z,des}, \omega_z = \omega_{z,des} = (2\pi/p)v_{z,des}$

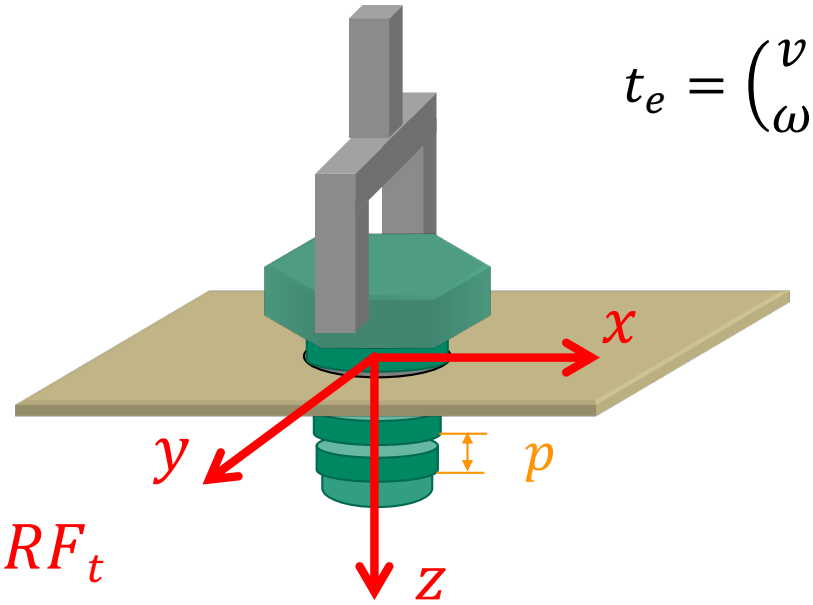
$f_z = f_{z,des}, \mu_z = \mu_{z,des}$ (one function of the other!)

the screw proceeds along and around the z-axis, but not in an independent way! (1-dof)

accordingly, f_z and μ_z cannot be independent

environment wrench direction should be orthogonal to robot motion twist!

Selection of directions – example 3



$$t_e = \begin{pmatrix} v \\ \omega \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 & \frac{2\pi}{p} \end{pmatrix}^T \quad v_z = S_t\, v \qquad (6-k=1)$$

or $\omega_z = 2\pi \dfrac{v_z}{p}$

S_w: such that $S_w^T\, S_t = 0$

$$f_z = -\frac{2\pi}{p}\mu_z \qquad (k=5)$$

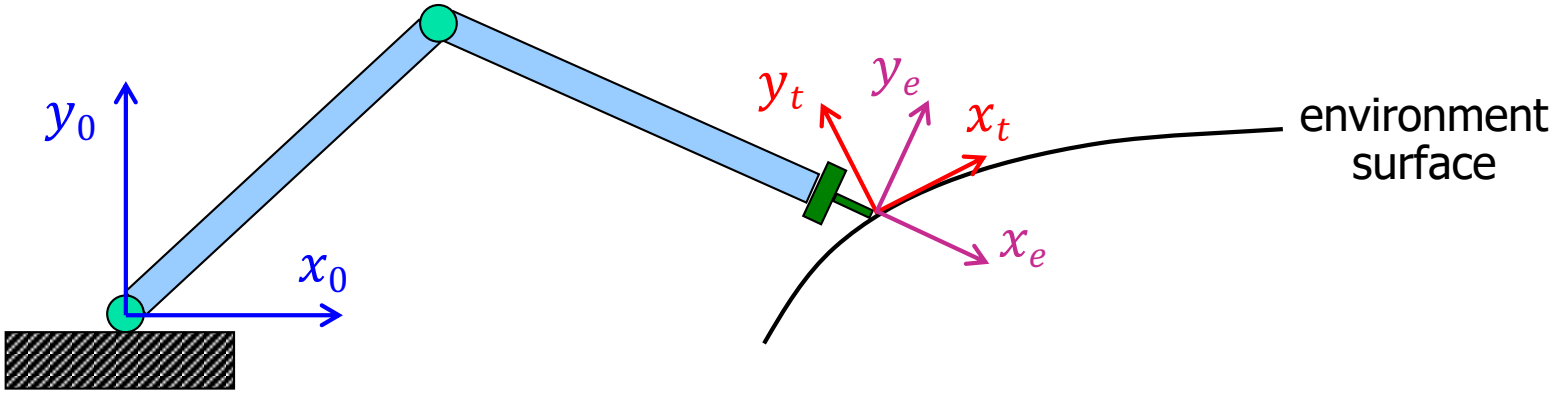
the columns of S_t and S_w no longer coincide with selected columns of the 6×6 identity matrix

⇒ screw directions

$$w_{\text{env}} = \begin{pmatrix} f \\ \mu \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -2\pi/p \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} f_x \\ f_y \\ \mu_x \\ \mu_y \\ \mu_z \end{pmatrix} = S_w\, \lambda$$

Frames of interest – example 4

planar motion of a 2R robot ($n = 2$) in pointwise contact with a surface (task dimension $m = 2$)



- task frame RF_t used for an independent definition of the hybrid reference values (here: ${}^t v_{x,des}$ $[m-k=1]$ and ${}^t f_{y,des}$ $[k=1]$) as well as for computing the errors that drive the feedback control law
- sensor frame RF_e (here: RF_2) where the force ${}^e f = ({}^e f_x,\ {}^e f_y)$ is measured
- base frame RF_0 in which the end-effector velocity is expressed (here: ${}^0 v = ({}^0 v_x,\ {}^0 v_y)$ of O_2), computed using robot Jacobian and joint velocities

all quantities (and errors!) should be expressed ("rotated") in the same reference frame ⇒ the task frame!

General parametrization of hybrid tasks

a "description" of robot-environment type of interaction at the end-effector: it implicitly defines the task frame

$$t_e = \begin{pmatrix} v \\ \omega \end{pmatrix} = S_t(q)\,v \qquad v \in \mathbb{R}^{6-k}$$

parametrizes robot e-e free motion

$$w_{\text{env}} = \begin{pmatrix} f \\ \mu \end{pmatrix} = S_t(q)\,\lambda \qquad \lambda \in \mathbb{R}^{k}$$

parametrizes reaction wrenches on e-e

reaction wrenches do not perform work on robot e-e twists

$$S_w^T(q)\,S_t(q) = 0$$

directions of the axes of the task frame may depend on $y_e = k_e(q)$ (i.e., on robot e-e pose in the environment)

+

robot dynamics

$$M(q)\ddot{q} + c(q,\dot{q}) + g(q) = \tau - \tau_{\text{env}}$$

controlling wrenches acting on the environment! ($\tau_{\text{env}} = -\tau_{\text{int}}$)

robot kinematics

$$t_e = J(q)\dot{q}$$

duality

$$\tau = J^T(q)\,w_e$$

$$\tau_{\text{env}} = J^T(q)\,w_{\text{env}}$$

Hybrid force/velocity control



- control objective: to impose desired task evolutions to the parameters υ for motion and to the parameters λ for interaction

$$\upsilon \longrightarrow \upsilon_d(t) \qquad \lambda \longrightarrow \lambda_d(t)$$

- the control law is designed in two steps
  1. exact linearization and decoupling in task space by feedback

closed-loop model → $\begin{pmatrix} \dot{\upsilon} \\ \dot{\lambda} \end{pmatrix} = \begin{pmatrix} a_\upsilon \\ a_\lambda \end{pmatrix}$

  2. (linear) design of a_υ and a_λ so to impose the desired dynamic behavior to the errors $e_\upsilon = \upsilon_d - \upsilon$ (and its integral) and $e_\lambda = \lambda_d - \lambda$

- assumption: $n = m = 6$ and $J(q)$ out of singularities

Note: in simple cases, υ and λ drive single components of v or ω and of f or μ; accordingly, S_t and S_w are columns of 0/1 selection matrices

Feedback linearization in task space
whiteboard …



- differentiate e-e twist t_e, using kinematics and free motion parametrization

$$\dot{t}_e = J(q)\ddot{q} + \dot{J}(q)\dot{q} = S_t(q)\dot{v} + \dot{S}_t(q)v$$

- substitute $\ddot{q}$ from this in the dynamic model, and then also $\dot{t}_e$

$$M(q)J^{-1}(q)\dot{t}_e - M(q)J^{-1}(q)\dot{J}(q)\dot{q} + c(q,\dot{q}) + g(q)$$

$$= M(q)J^{-1}(q)S_t(q)\dot{v} + M(q)J^{-1}(q)\left(\dot{S}_t(q)v - \dot{J}(q)\dot{q}\right) + c(q,\dot{q}) + g(q) = \tau - \tau_{\text{env}}$$

- pre-multiply by J^{-T}, dropping here dependences on $(q,\dot{q})$

$$\underbrace{J^{-T}M\,J^{-1}}_{M_e(q)}S_t\dot{v} + \underbrace{J^{-T}MJ^{-1}\left(\dot{S}_t v - \dot{J}\dot{q}\right) + J^{-T}(c+g)}_{n_e(q,\dot{q})} = J^{-T}(\tau - \tau_{\text{env}}) = w_e - w_{\text{env}}$$

$M_e(q)$: task-space robot inertia

… and use the contact wrench parametrization

$$M_e(q)S_t(q)\dot{v} = w_e - n_e(q,\dot{q}) - S_w(q)\lambda \qquad (*)$$

- pre-multiplying $(*)$ by S_t^T, eliminate the presence of λ since $S_t^T S_w = 0$

$$\underbrace{S_t^T(q)M_e(q)S_t(q)}_{\text{nonsingular } (6-k)\times(6-k)}\dot{v} = S_t^T(q)(\,\underbrace{w_e}_{\text{control in task space}} - n_e(q,\dot{q}))$$

Feedback linearization in task space

whiteboard …



- using the pseudoinverse of S_w weighted by the task-space robot inertia M_e

$$(S_w(q))^{\#}_{M_e} = (S_w^T(q)M_e^{-1}(q)S_w(q))^{-1}S_w^T(q)M_e^{-1}(q)$$

pre-multiply $(*)$ so to isolate the wrench parameter λ

$$\underbrace{(S_w(q))^{\#}_{M_e}M_e(q)S_t(q)}\,\dot{v} = (S_w(q))^{\#}_{M_e}(w_e - n_e(q,\dot{q})) - \underbrace{(S_w(q))^{\#}_{M_e}S_w(q)}\lambda$$

vanishes since $S_w^T\,S_t = 0$ $\qquad$ $= I$

- so far, for motion twists and contact wrenches in task space we have

$$S_t^T(q)M_e(q)S_t(q)\dot{v} = S_t^T(q)(w_e - n_e(q,\dot{q})) \qquad \lambda = (S_w(q))^{\#}_{M_e}(w_e - n_e(q,\dot{q}))$$

- thus, defining the control law

$$w_e = n_e(q,\dot{q}) + M_e(q)S_t(q)a_v + S_w(q)a_\lambda \qquad (**)$$

yields finally the linear and decoupled subsystems

$$\dot{v} = a_v \;\}\; 6-k \qquad \lambda = a_\lambda \;\}\; k$$

end of step 1

- $(**)$ can be rewritten in terms of joint torque $(\tau = J^T(q)w_e)$

$$\tau = M(q)J^{-1}(q)\left(S_t(q)a_v + \dot{S}_t(q)v - \dot{J}(q)\dot{q}\right) + J^T(q)S_w(q)a_\lambda + c(q,\dot{q}) + g(q)$$

Stabilization with a_v and a_λ



- for step 2, we apply as usual linear control techniques for the (exponential) stabilization of the errors on each decoupled channel
- k scalar force/torque control loops: $e_\lambda = \lambda_d - \lambda \longrightarrow 0$

$$a_\lambda = \lambda_d$$

would be enough, since no dynamics is assumed in constrained task directions

$$a_\lambda = \lambda_d + K_I \int (\lambda_d - \lambda) dt$$

adding an integral error gives robustness (e.g., against constant disturbances)

$K_I \geq 0$ diagonal

- $6 - k$ scalar velocity control loops: $e_v = v_d - v \longrightarrow 0$

$$a_v = \dot{v}_d + K_{Pv}(v_d - v) + K_{Iv} \int (v_d - v) dt$$

$K_{Pv}, K_{Iv} > 0$ diagonal

- if explicit e-e constraints $f(y_e) = f\big(k_e(q)\big) = 0$ are also available, one can extract from these a reduced set of $6 - k$ free position variables $r = r(q)$, with associated scalar position control loops: $e_r = r_d - r \longrightarrow 0$

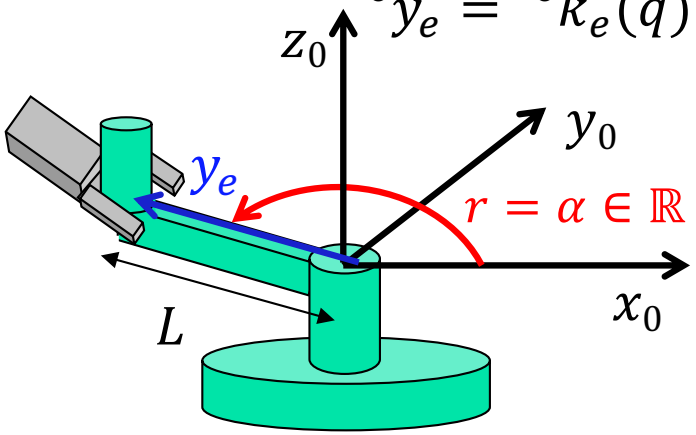
$$a_v = \ddot{r}_d + K_{Dr}(\dot{r}_d - v) + K_{Pr}(r_d - r(q))$$

$K_{Pr}, K_{Dr} > 0$ diagonal

"Filtering" position and force measures

v and r are obtained from measures of q and $\dot{q}$, equating e-e pose and velocity from robot side (robot kinematics) and from task/environment side

example



$$ {}^0y_e = {}^0k_e(q) = \begin{pmatrix} L\cos r \\ L\sin r \\ 0 \end{pmatrix} \Rightarrow r = \text{atan2}\left({}^0k_{ey}(q),\ {}^0k_{ex}(q) \right) = r(q) $$

$$ J(q)\dot{q} = S_t(q)v \Rightarrow v = S_t^{\#}(q)J(q)\dot{q} $$

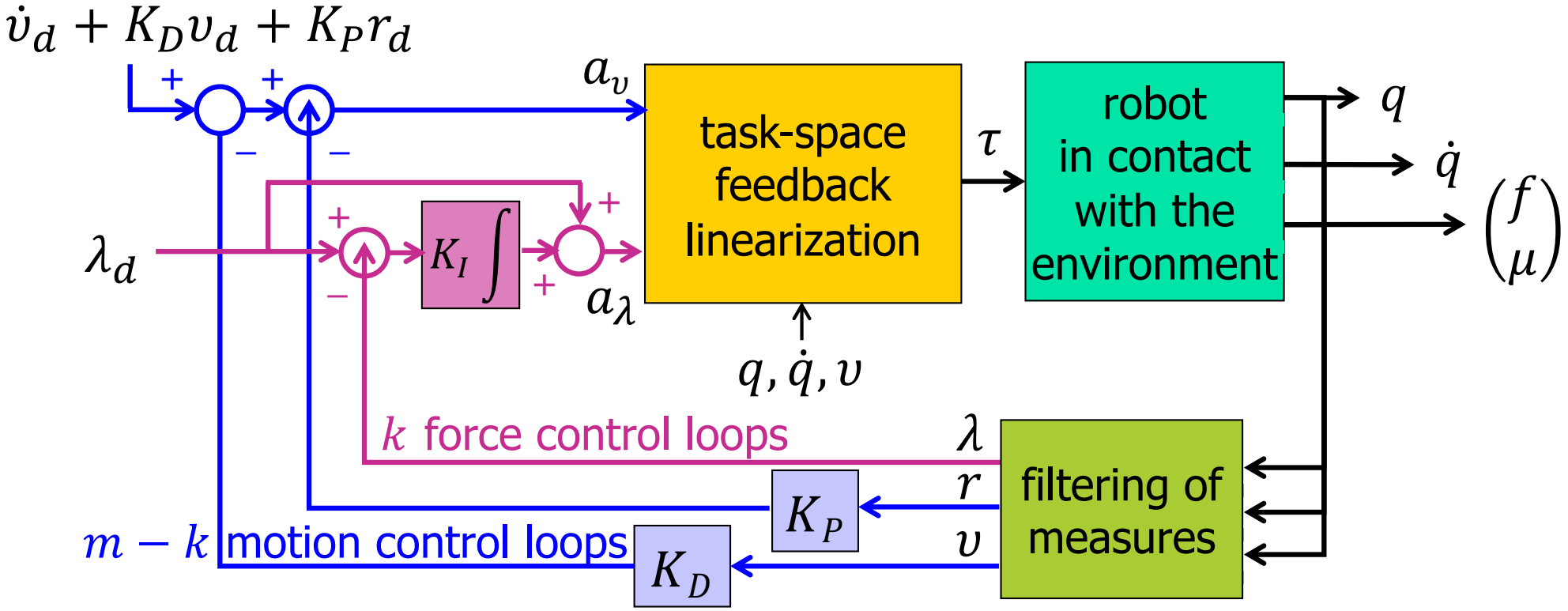
pseudoinverse of "tall" matrix with full column rank

$$ S_t^{\#} = (S_t^T S_t)^{-1} S_t^T $$

similarly, λ is obtained from force/torque measures at the end-effector

$$ \begin{pmatrix} f \\ m \end{pmatrix} = S_w(q)\lambda \Rightarrow \lambda = S_w^{\#}(q) \begin{pmatrix} f \\ m \end{pmatrix} $$

Block diagram of hybrid control



references $\lambda_d = S_w^{\#}(q) w_d$ and $v_d = S_t^{\#}(q) t_d$ (from desired e-e wrench and twist)

usually $m = 6$ (complete 3D space)

limit cases $k = 0$: no force control loops, only motion (motion in free space)

$k = m$: no motion control loops, only force ("frozen" robot end-effector)

Block diagram of hybrid control

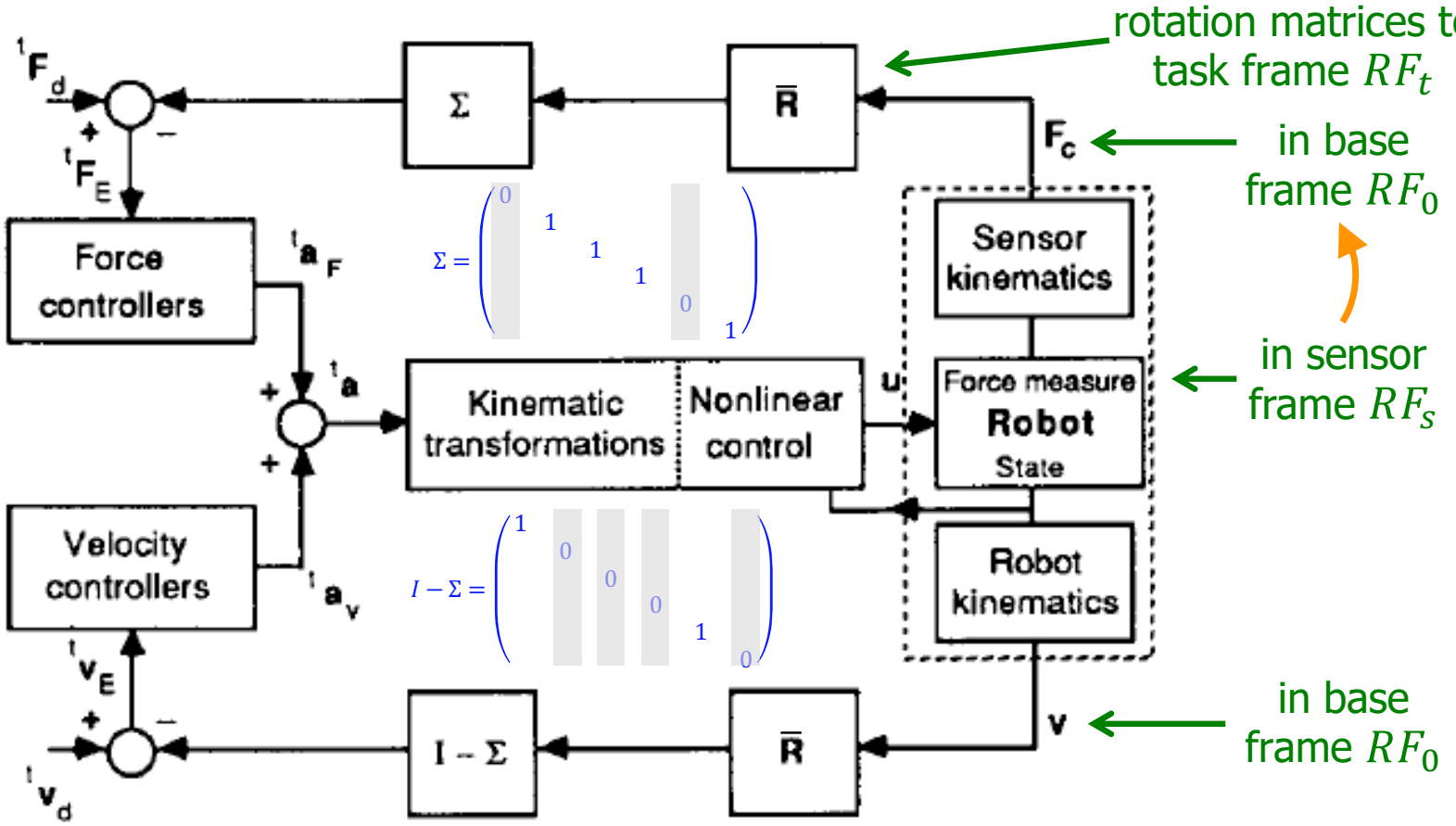
classical case of 0/1 selection matrices

notation in this slide

$$F = \begin{pmatrix} f \\ \mu \end{pmatrix}$$

$$V = \begin{pmatrix} v \\ \omega \end{pmatrix}$$

$$\bar{R} = \begin{pmatrix} R & 0 \\ 0 & R \end{pmatrix}$$



λ and v are just single components of f (or μ) and v (or ω)

S_λ and S_v are columns of 0/1 selection matrices: Σ and $I - \Sigma$

illustrated here for example 1, slide #5 (discarding **zero** columns to get S_λ and S_v)

Force control via an impedance model



- in a force-controlled direction of the hybrid task space, when the contact stiffness is limited (i.e., far from infinite, as assumed in the ideal case), one may use impedance model ideas to explicitly control the contact force
  - let x be the position of the robot along such a direction, x_d the (constant) contact point, $k_s > 0$ the contact (or the sensor) stiffness, and $f_d > 0$ the desired contact force
- the impedance model is chosen then as

$$m_m \ddot{x} + d_m \dot{x} + k_s (x - x_d) = f_d$$

  where the force sensor measures $f_s = k_s (x - x_d)$, and only $m_m > 0$ and $d_m > 0$ are free model parameters
- after feedback linearization ($\ddot{x} = a_x$), the command a_x is designed as

$$a_x = (1/m_m)[(f_d - f_s) - d_m \dot{x}]$$

  which is a P-regulator of the desired force, with velocity damping
- the same control law works also before the contact ($f_s = 0$), guaranteeing a steady-state speed $\dot{x}_{ss} = f_d / d_m > 0$ in the approaching phase

First experiments with hybrid control







video

video



MIMO-CRF robot
(DIS, Laboratorio di Robotica, 1991!!)

Sources of inconsistency
in force and velocity measurements



1. presence of friction at the contact
   - ➜ a reaction force component appears that opposes motion in a "free" motion direction (in case of Coulomb friction, the tangent force intensity depends also on the applied normal force ...)
2. compliance in the robot structure and/or at the contact
   - ➜ a (small) displacement may be present also along directions that are nominally "constrained" by the environment

NOTE: if the environment geometry at the contact is perfectly known, the task inconsistencies due to 1. and 2. on parameters υ and λ are already filtered out by the pseudoinversion of matrices S_υ and S_λ

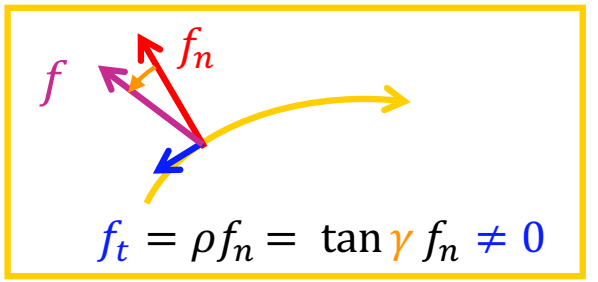
3. uncertainty on environment geometry at the contact
   - ➜ can be reduced/eliminated by real-time estimation processes driven by external sensors (e.g., vision –but force too!)

Estimation of an unknown surface

how difficult is to estimate the unknown profile of the environment surface, using information from velocity and force measures at the contact?

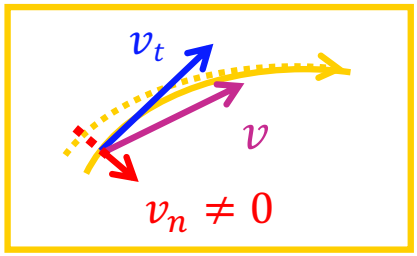
1. normal = nominal direction of measured force

   ... in the presence of contact motion with friction, the measured force f is slightly rotated from the actual normal by an (unknown) angle γ



2. tangent = nominal direction of measured velocity

   ... compliance in the robot structure (joints) and/or at the contact may lead to a computed velocity v having a small component along the actual normal to the surface

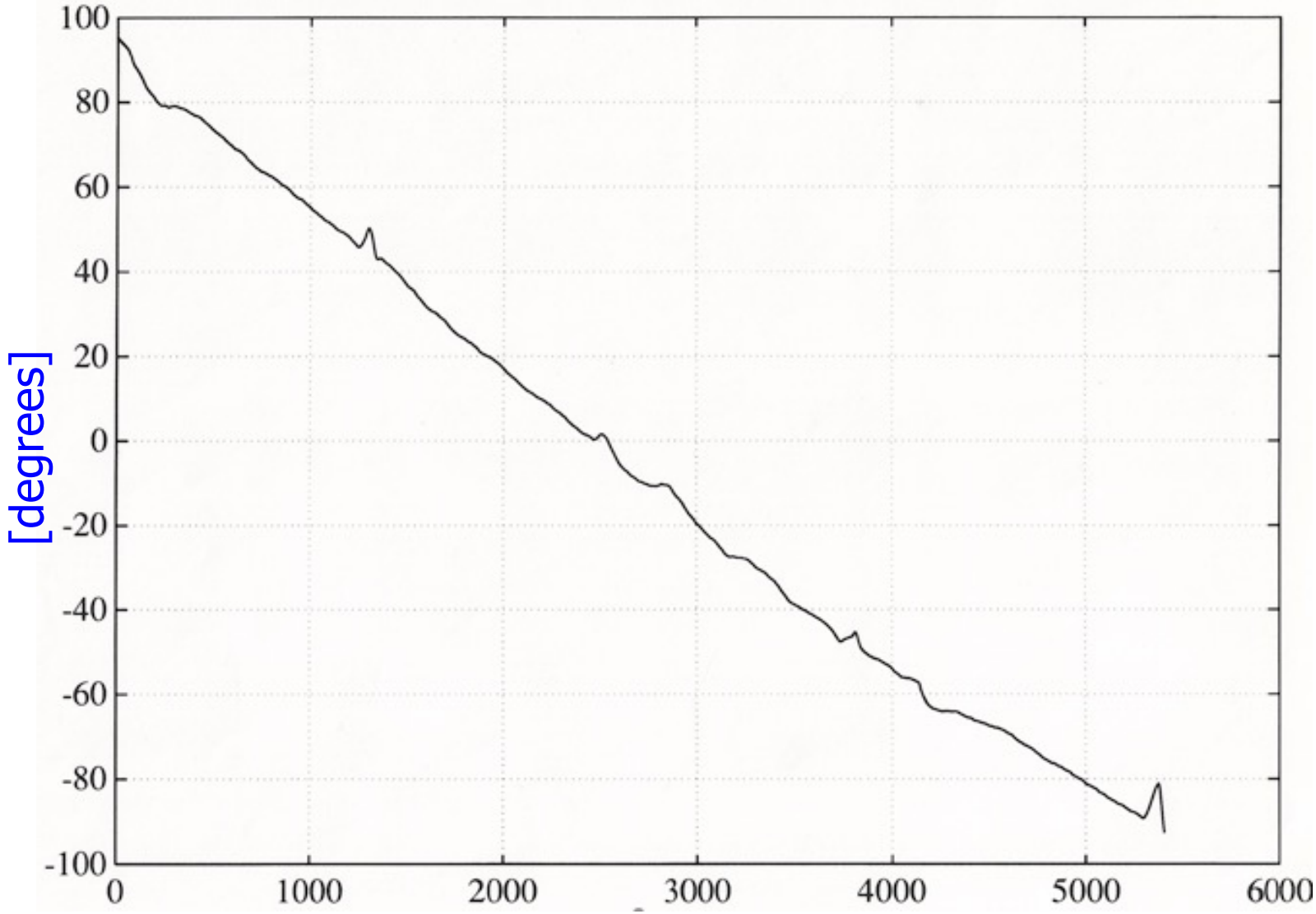


3. mixed method (sensor fusion) with RLS
   a. tangent direction is estimated by a Recursive Least Squares method from position measurements
   b. friction angle is also estimated by a Recursive Least Squares method, using the current estimate of the tangent direction and from force measurements

to approach an unknown surface or to recover contact (in case of loss), the robot uses simple exploratory moves

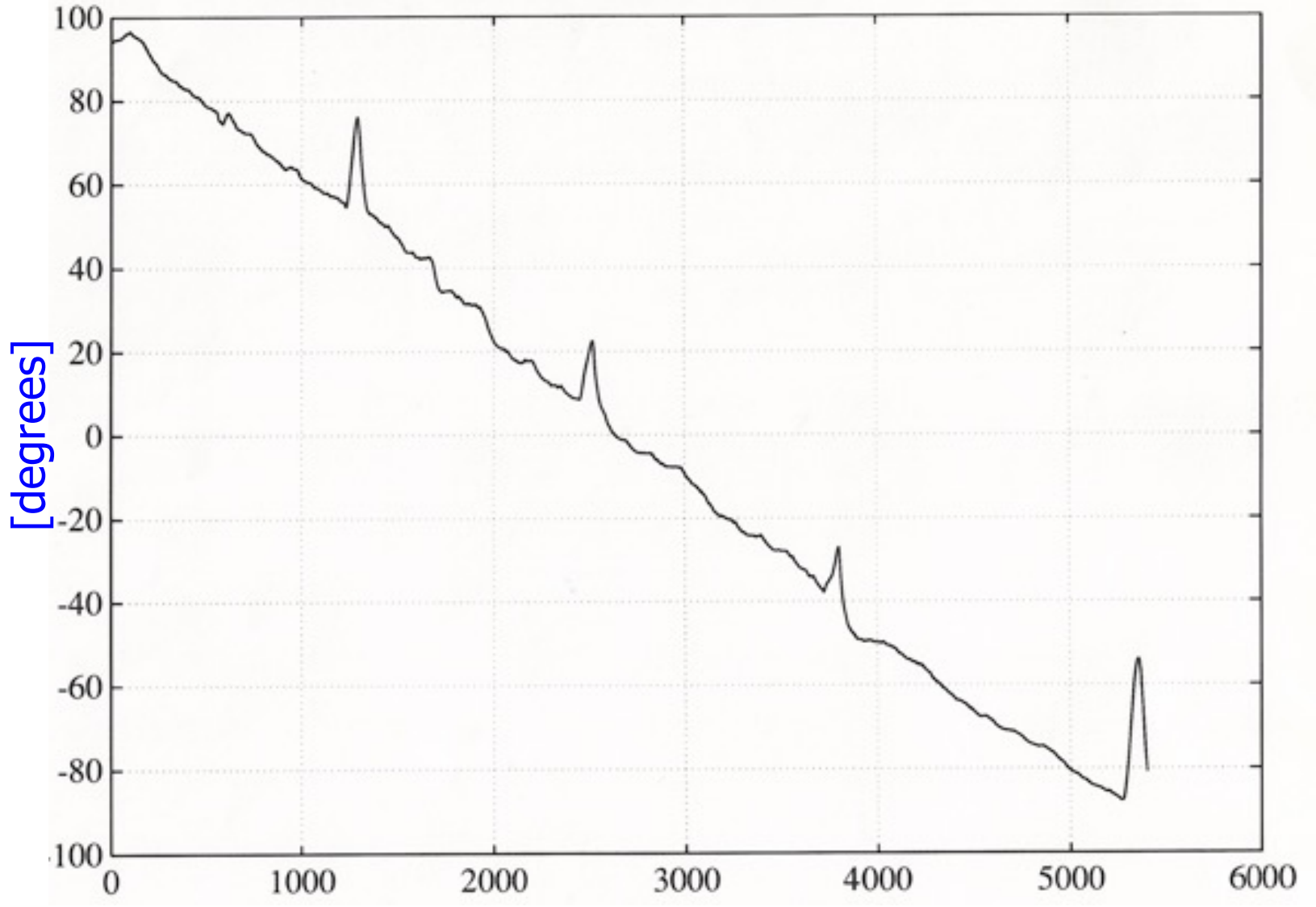
Position-based estimation of the tangent

(for a circular surface traced at constant speed)

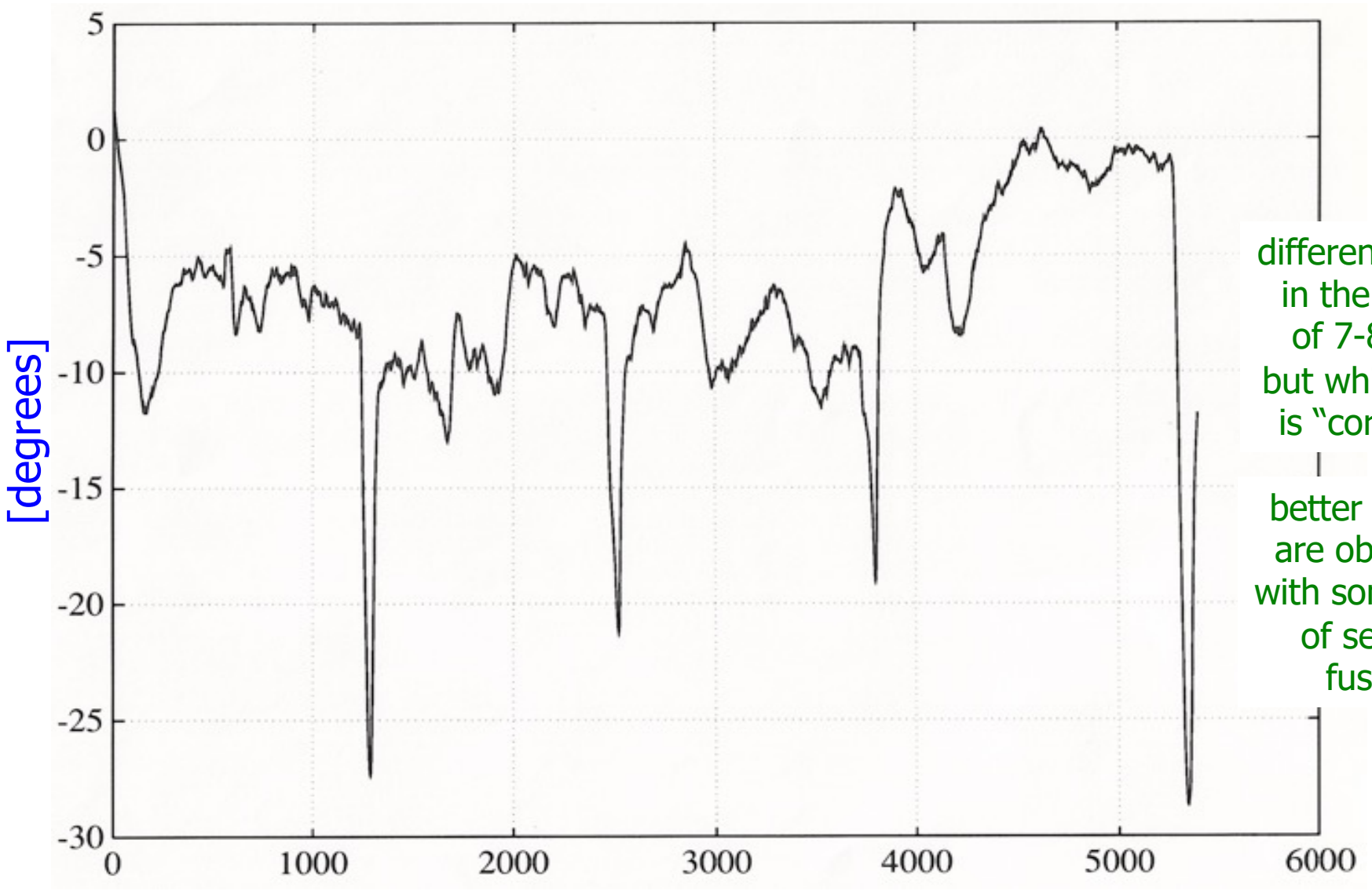


Force-based estimation of the tangent

(for the same circular surface traced at constant speed)



Difference between estimated tangents

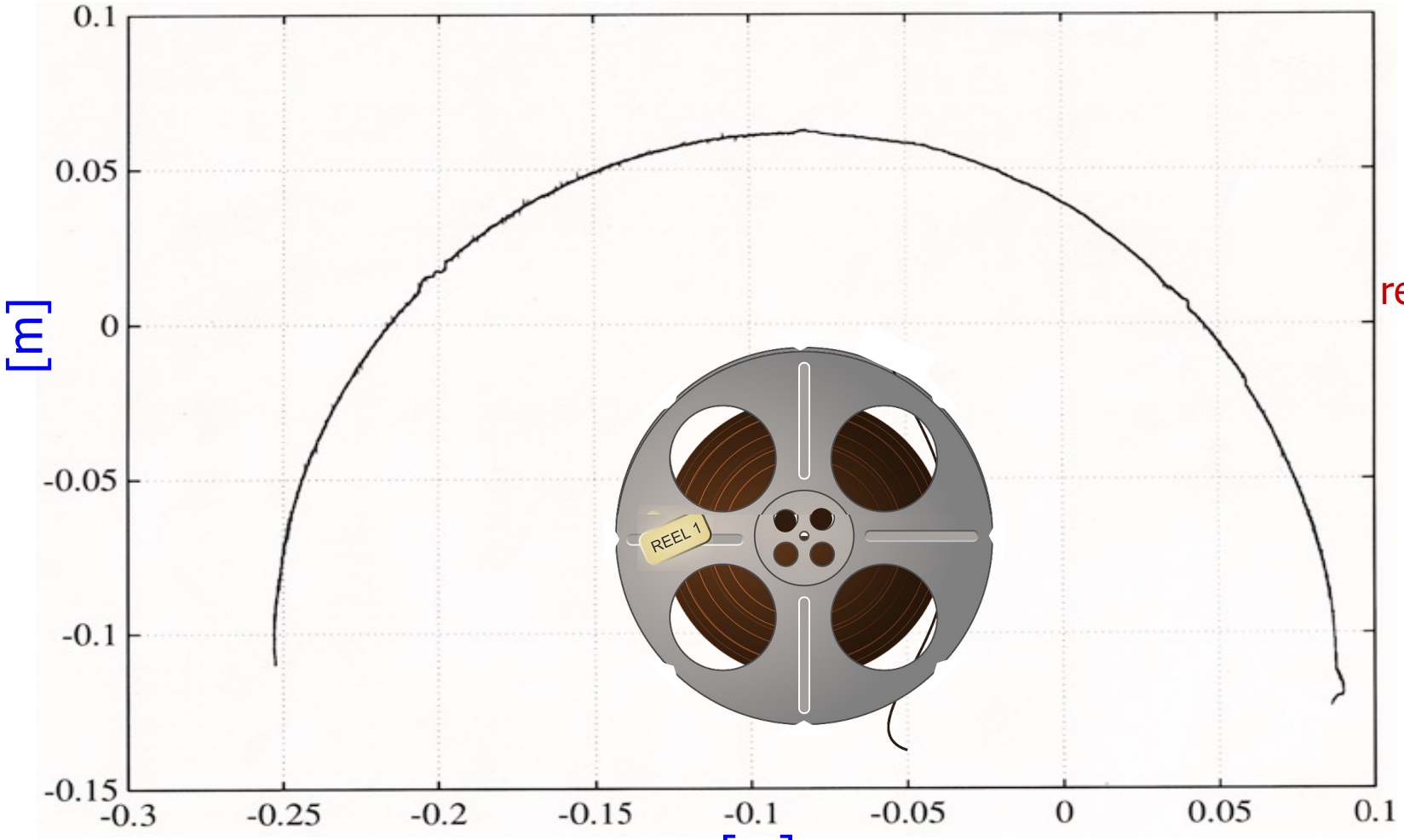


differences are in the order of 7-8° ... but which one is "correct"?

better results are obtained with some kind of sensor fusion

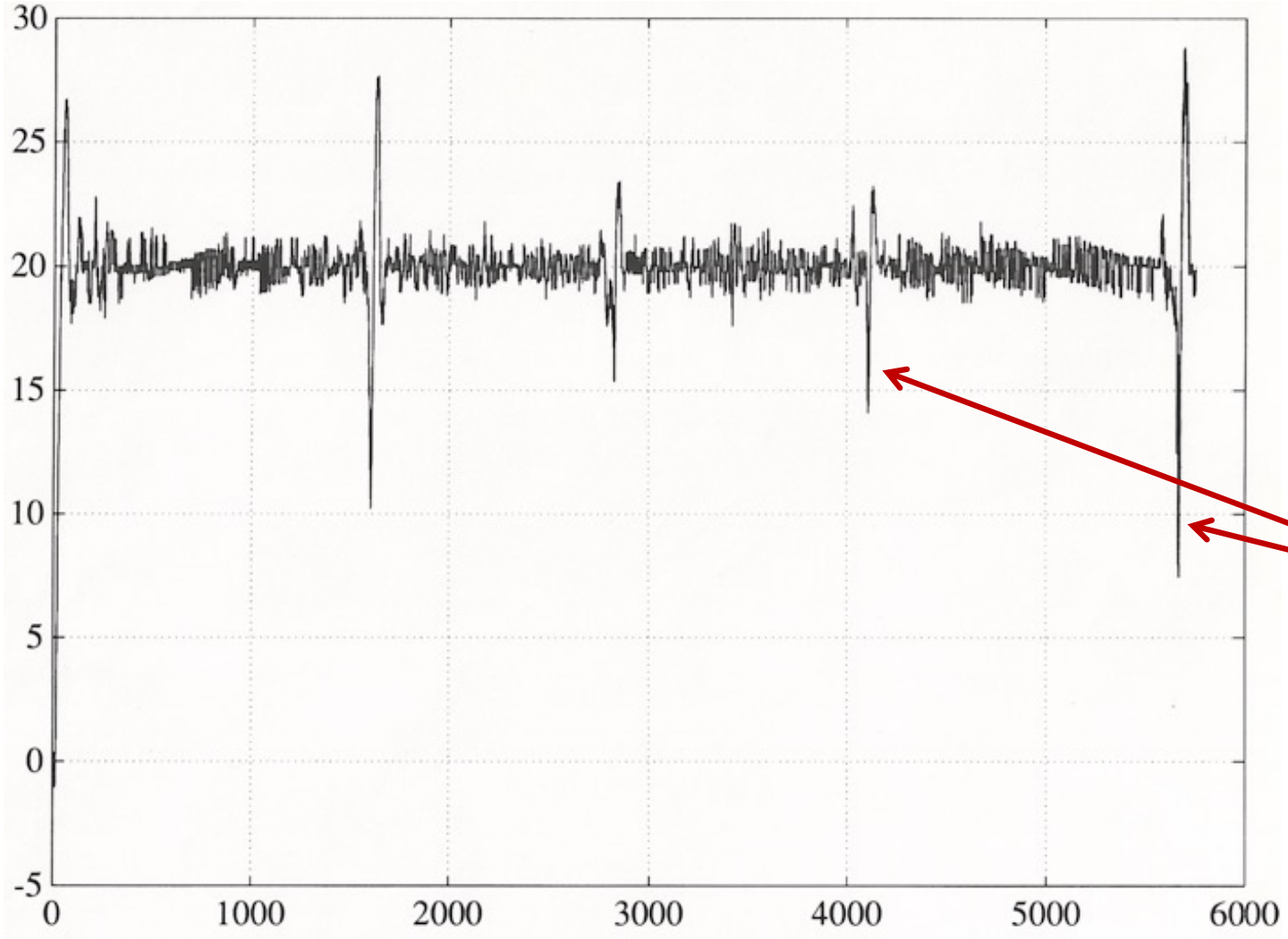
Reconstructed surface profile

estimation by a RLS (Recursive Least Squares) method: we continuously update the coefficients of two quadratic polynomials that fit locally the unknown contour, using data fusion from both force and position/velocity measurements



this is the reconstructed contour of a cinema "film reel" (of radius = 17 cm)

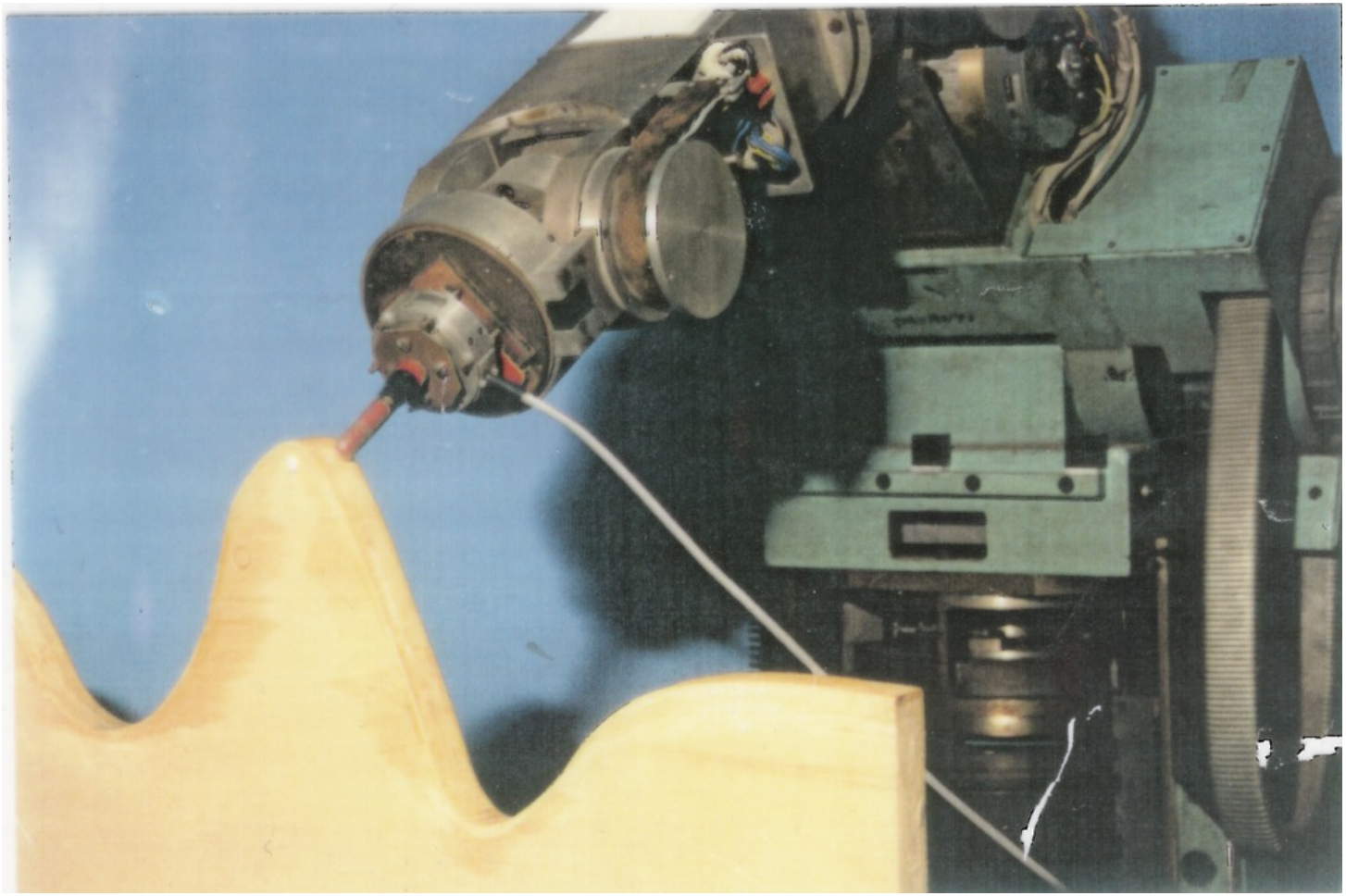
Normal force



regulated to 20 N during simultaneous motion and estimation

force peaks correspond to the grooves on the surface contour

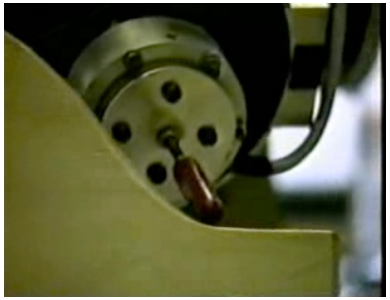
Contour estimation and hybrid control
performed simultaneously



MIMO-CRF robot (DIS, Laboratorio di Robotica, 1992)

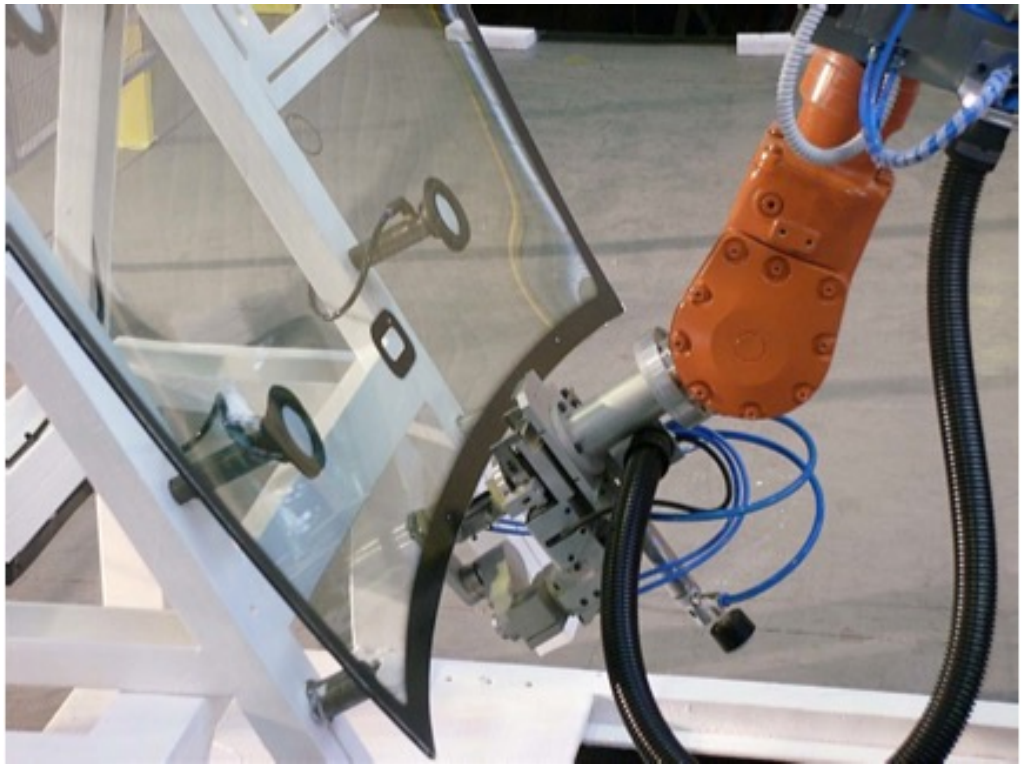
Contour estimation and hybrid control





video

Robotized deburring of car windshields



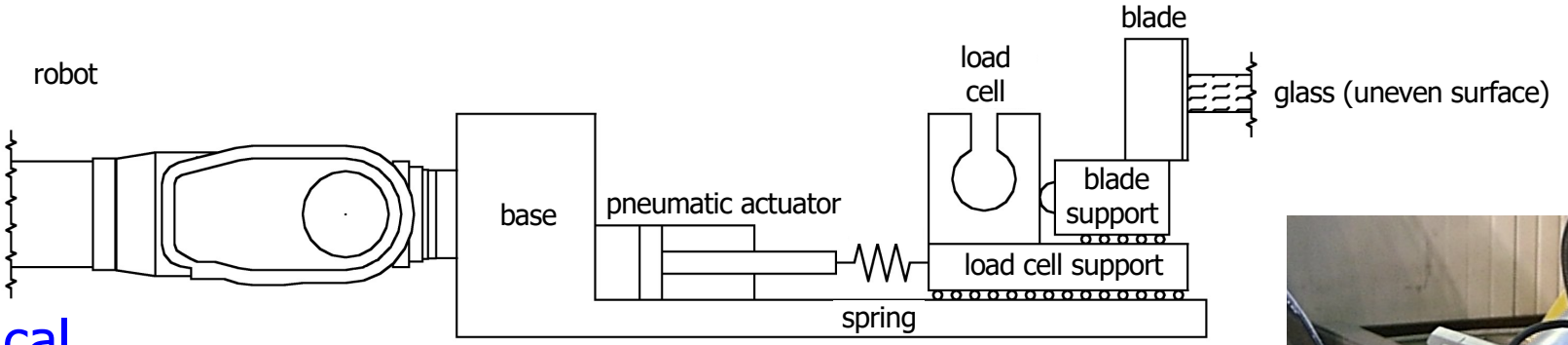
- car windshield with sharp edges and fabrication tolerances, with excess of material (PVB = Polyvinyl butyral for gluing glass layers) on the contour
- robot end-effector follows the pre-programmed path, despite the small errors w.r.t. the nominal windshield profile, thanks to the compliance of the deburring tool
- contact force between tool blades and workpiece can be independently controlled by a pneumatic actuator in the tool

the robotic deburring tool contains in particular

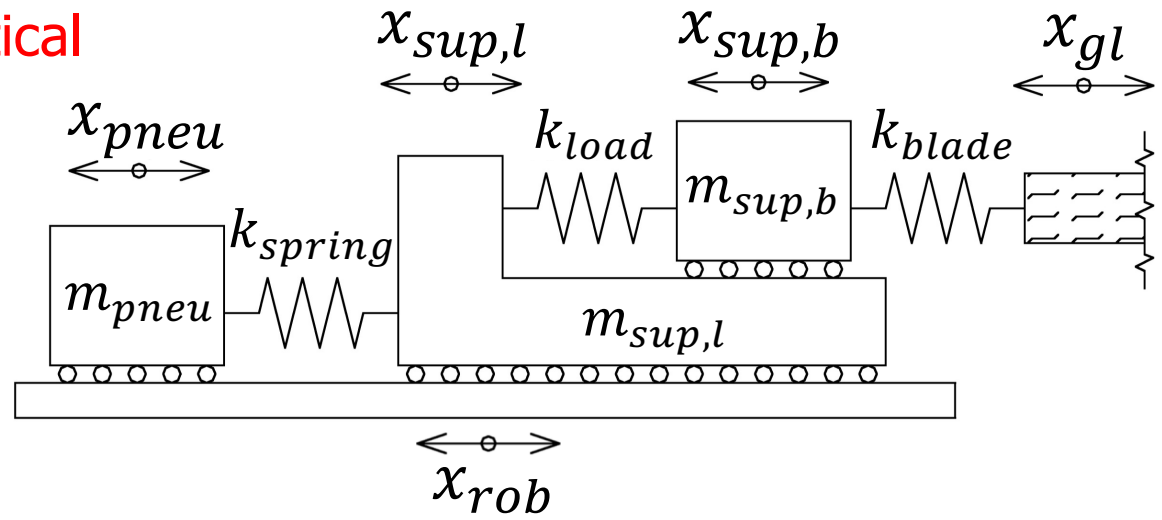
- two blades for cutting the exceeding plastic material (PVB), the first one actuated, the second passively pushed against the surface by a spring
- a load cell for measuring the 1D applied normal force at the contact
- on-board control system, exchanging data with the ABB robot controller

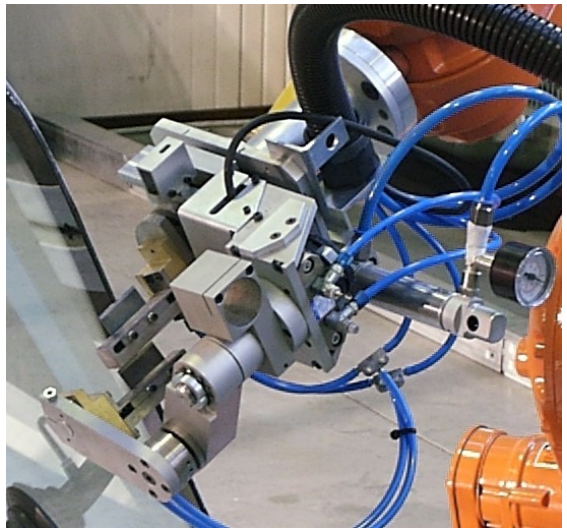
Model of the deburring work tool

physical



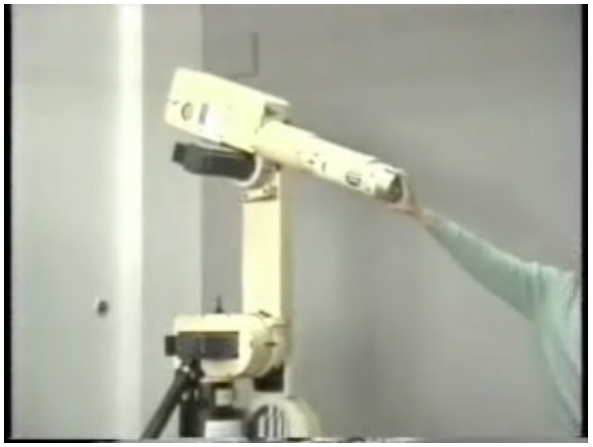
mathematical



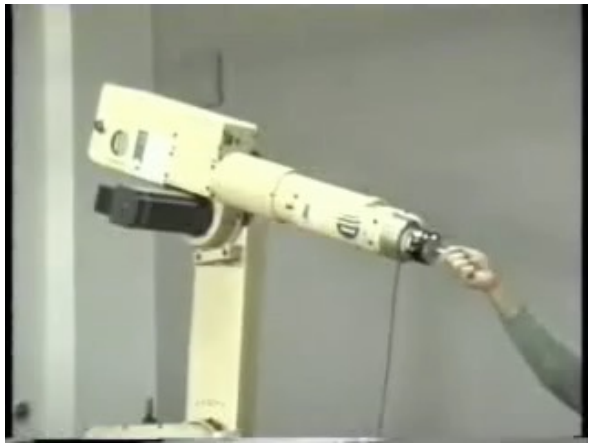


for stability analysis (based on linear models and root locus techniques)
of force control in a single direction and in presence of multiple masses/springs,
see again Eppinger & Seering, IEEE CSM, 1987 in material the course website

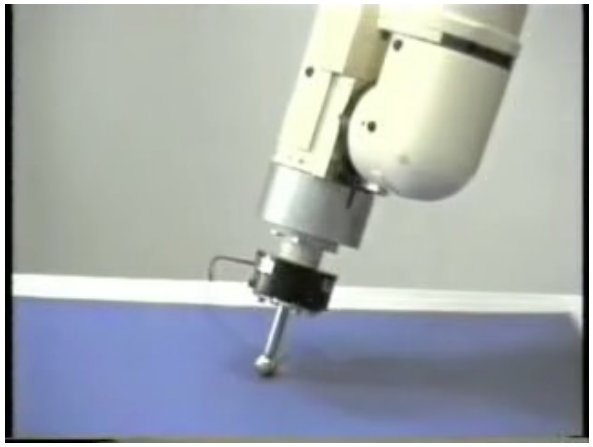
Summary through video segments



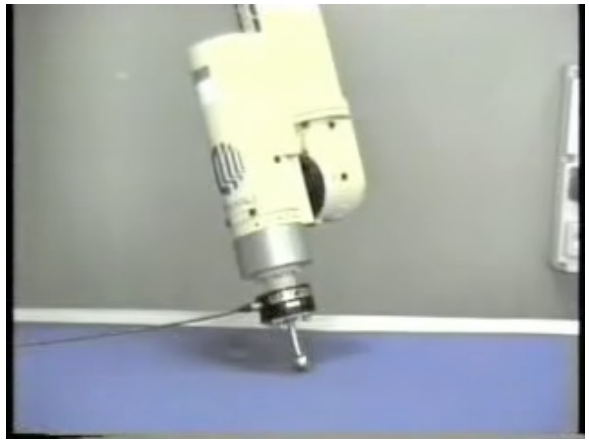
compliance control
(active Cartesian stiffness control without F/T sensor)



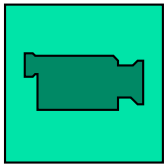
impedance control
(with F/T sensor)



force control
(realized as external loop providing the reference to an internal motion loop on position or velocity –see Appendix)



hybrid force/position control



COMAU Smart robot
c/o Università di Napoli, 1994

(full video on course web site)

Appendix

- force control can also be realized as an external loop providing reference values to an internal motion loop, here of position type
- inner-outer (or cascaded) control scheme

(see video in slide #33)

  - angular position quantities (e-e orientation, errors, commands) can be expressed in different ways (Euler angles $\boldsymbol{\phi}$, rotation matrices $\boldsymbol{R}$, ...)

